

Nathan James Anderson

Tulsa, OK | anderson.nathanjames@gmail.com | (704) 604-4165 | [linkedin.com/in/njand](https://www.linkedin.com/in/njand) | github.com/njand | njand.github.io

Applied NLP scientist specializing in multilingual speech and language assessment, with models built and evaluated across 70+ languages. M.S. in Language Technologies, Carnegie Mellon University.

EXPERIENCE

The Church of Jesus Christ of Latter-day Saints

Provo, UT (Remote)

Product Manager (NLP Research Lead)

Oct 2022 – Present

- Built a multilingual oral proficiency scoring system for 9 languages (Whisper, spaCy, mmBERT + LoRA, 3-stage transfer learning) to replace a \$300k/year vendor; in a 3-language gold-standard evaluation with professionally trained raters, raised agreement from **0.36 to 0.86** (quadratic weighted κ), with **conservatively projected savings of ~\$225k/year**.
- Prototyped the scoring model independently, directed 8 research assistants through dozens of architecture and feature experiments, and built the deployment pipeline on Azure ML.
- Engineered LLM-based grammatical error correction feedback adapted to learner proficiency (few-shot prompting, structured outputs, OpenAI API) with automated and human-in-the-loop evaluation; deployed in **41 languages for 5,000+ learners annually**.
- Mined multimillion-sentence corpora across 70+ languages for pedagogical examples (spaCy parsing, parallelized simplicity heuristics) and fine-tuned a BERT safety classifier, leading annotation rounds to maintain high inter-annotator agreement.
- Developed a beta conversational practice chatbot with STT/TTS, safety guardrails, and generated personas designed to avoid cultural stereotyping, validated in 10+ languages.

Lowe's

Mooresville, NC (Remote)

Data Science Graduate Intern

Jun 2021 – Aug 2021

- Designed a Siamese neural network (TensorFlow), with K-means clustering to build training data, scoring attribute similarity across 3,000+ catalog features for product recommendations; received a full-time return offer.

The Church of Jesus Christ of Latter-day Saints

Riverton, UT

Software Development Intern (NLP Research)

Jul 2019 – Dec 2019

- Developed an English-to-38-language speech-to-speech translation prototype integrating custom ASR, MT, and TTS; fine-tuned Azure Speech-to-Text on 50+ hours of curated audio, reducing **WER from 7.2% to 3.3%**.
- Designed a low-latency dynamic programming algorithm aligning streaming ASR output to scripted transcripts via phonetic matching to reduce live-translation latency (published at AMTA 2022).

PUBLICATIONS & PROJECTS

- Latin ASR:** Hand-transcribed 10 hours of Latin speech; fine-tuned wav2vec2 on Modal GPUs; Optuna-tuned KenLM decoding cut WER from 13.2% to 8.7% (5-fold CV); ONNX INT8 export (72% smaller, ~3x faster on CPU); GitHub Actions CI; released Latin BERT normalization model (PyPI) and 11M-token dataset: huggingface.co/spaces/njand/latin-asr-demo
- Anderson, N., Wilson, C., & Richardson, S. D. (2022).** *Lingua: Addressing Scenarios for Live Interpretation and Automatic Dubbing*. AMTA, Users and Providers Track.
- 3rd place, SIGMORPHON 2021 shared task on predicting native-speaker plausibility judgments of inflected nonce words.

SKILLS

ML & NLP: Python, PyTorch, Hugging Face (Transformers, PEFT/LoRA), spaCy, Stanza, scikit-learn, XGBoost, OpenAI API, pandas, NumPy

Speech: Whisper/WhisperX, wav2vec2, KenLM/pyctcdecode, pyannote.audio, Praat, Azure Speech, jiwer (WER/CER)

Tools: Git, GitHub Actions, SQL, Bash/Linux, Azure ML, Modal, ONNX Runtime, Flask, Gradio, Jira

Languages: English (native); Portuguese, Spanish, Latin (advanced); Indonesian, Turkish (intermediate)

EDUCATION

Carnegie Mellon University — M.S., Language Technologies (GPA 3.90)

Pittsburgh, PA | Aug 2022

Research: low-resource morphology (Nahuatl, Mixtec), cognate identification, semantic taxonomy acquisition in language models

Brigham Young University — B.A., Linguistics, cum laude (GPA 3.93)

Provo, UT | Apr 2020

Minors in Computer Science and Linguistic Computing. Full-tuition scholarship; FLAS Fellowships (Indonesian, Portuguese); Boren Scholarship (Turkish)